Density Coverage-Based Exemplar Selection for Incremental SAR Automatic Target Recognition

Bin Li, Zongyong Cui, *Member, IEEE*, Yuxuan Sun, Jianyu Yang, *Member, IEEE*, and Zongjie Cao, *Member, IEEE*

***Abstract*— The traditional synthetic aperture radar automatic target recognition (SAR/ATR) algorithm can train a sufficient number of known class samples and classify the samples in the test set. However, if the old model is trained only with the new class samples, the old class samples' knowledge is easily forgotten by the new model, which is called catastrophic forgetting. The reason is that the model only fits the distribution of current training samples, so training the whole dataset is necessary. Due to the limitation of storage resources, it is often not feasible to retain the whole dataset. In order to avoid this phenomenon, a small number of old class samples can be kept to train with the new class samples. Therefore, how to select the old class samples becomes the key point. In this article, the density coverage-based exemplar selection (DCBES) is proposed to choose the key samples of the old class. DCBES selects samples based on the metric learning theory and the set covering theory. First, the metric learning theory is used to measure the similarity between samples and to obtain the density range of samples. Then, the exemplar selection problem is considered a set covering problem, to select a fixed number of exemplars to achieve the maximum coverage of the class density range. Experiments on the moving and stationary target acquisition and recognition (MSTAR) dataset show that our method is superior to other exemplar selection methods and achieves the best results.**

***Index Terms*— Density region, exemplar selection, greedy algorithm, incremental learning.**

I. Introduction

DEEP learning promotes the rapid development of various fields in synthetic aperture radar (SAR), especially SAR automatic target recognition (ATR) [1], [2], [5], [21], [28], [51]. It is very important for SAR ATR to select discriminative features. Liu et al. [28] introduced a dual-channel feature map convolutional neural network (DCFM-CNN) to extract features for SAR ATR. Amrani and Jiang [1] proposed a feature extraction method by exploiting the extracted deep features from CNNs on SAR images to introduce more powerful discriminative features. Guo et al. [21] developed a CNN extension based on Hinton's capsule network for SAR ATR. Amrani et al. [5] showed a feature extraction and classification algorithm based on a visual saliency model, which can effectively extract features. Zhang et al. [51] proposed a method that fused the scattering center feature and the deep CNN feature. Amrani et al. [2] introduced a novel feature extraction method based on bag of visual words (BoVW). The methods mentioned above effectively solve the problem of SAR image feature extraction.

Feature extraction plays an important role not only in image target recognition but also in image target detection [4], [6], [20], [31]. Gao et al. [20] showed a novel split convolution block (SCB) to enhance the feature extraction of small targets. Lv et al. [31] developed a method by using a stacked contractive autoencoder (sCAE) to extract the temporal change feature from the super-pixel with noise suppression. Amrani et al. [6] proposed a robust feature extraction method for SAR image target detection by adaptively fusing effective features from different CNN layers. Amrani et al. [4] proposed a very deep CNN (VDCNN) for abnormal electrocardiogram (ECG) detection. The methods mentioned above effectively prove the role of feature extraction in image target detection.

The current pattern of SAR ATR is mainly trained off-line with fixed training data, which brings a certain problem. Specifically, when a model has been trained through the old class samples, if some new class samples arrive, the model must be retrained with all the samples (old and new) to identify all classes of the test set. The reason for retraining is that the deep neural network only fits the distributions of current training data. If the model is trained with only new class samples, the classification boundary of the new model with respect to the old class will change sharply, resulting in the catastrophic forgetting [33] of the old class knowledge.

In actual application scenarios, the data acquired by sensors should come in the form of streams, and different types of data in different time periods will be acquired. Although deep learning has made great progress in the field of SAR ATR, the working mode will no longer be applicable if the data come in the form of stream, due to its off-line pattern. Obviously, this training mode is very inefficient, and massive resources will be wasted by the conservation of the original old class data. In addition, the rapid updating of the model is also impossible with the old model. To solve this problem, a series of incremental learning methods was proposed in SAR ATR [16], [19], [26], [29], [30], [41], [52]. A class separation loss function was designed in [52] to reduce the intraclass distance and increase the interclass distance. The high plastic

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The authors are with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: zjcao@uestc.edu.cn).

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error correction incremental learning (HPecIL) was proposed in [41] to address the model degradation and plasticity decline in the incremental scenario. An open set model with incremental learning (OSmIL) method was introduced in [16], which can learn the new unknown classes continually. An improved attributed scattering model was introduced in [26] to mathematically unify the scattering models of several canonical primitives. A novel incremental Wishart broad learning system (IWBLS) was designed in [19] to achieve polarimetric SAR (PolSAR) image classification. Incremental learning is a learning method that can quickly update the model with new class samples and a small number of old class samples. A class-incremental SAR target identification approach based on continuous teacher–student learning was proposed in [30] to prevent the student model from catastrophic forgetting. An incremental multitask learning method based on structured pruning was introduced in [29] to acquire the processing capabilities for new tasks while maintaining the processing capabilities for old tasks. A novel incremental nonnegative matrix factorization (INMF) for SAR target recognition was shown in [15] to overcome the defect that conventional methods have in online processing. Incremental learning can retain most of the original old knowledge and prevent the catastrophic forgetting of the old model.

Considering that generative adversarial network (GAN) is an effective way to generate samples [7], [8], there are two major ways of old knowledge's preservation in incremental learning: The old class samples are memorized in the previous training stage by GAN or autoencoder [12], [13], [24], [50], and then, when the new class samples come, the old class samples generated by the stored memory will be trained with the new. The small number of old class samples (exemplars) will be conserved by an exemplar selection method directly [9], [34], [35], [45], and then, the new model is trained with the exemplars and new samples. For the generating method, the advantage is that there is no need to keep the real sample, which can prevent the data leakage and the waste of storage resources. The disadvantage of the sample generation method is not only that the quality of the generated samples is not as good as the original samples but also that the trained generation models need to be saved. For the method of exemplar conservation, the superiority is that the old samples are kept directly, which yet causes leakage and waste. In the practical application of SAR ATR, the conditions for saving real samples are mainly reflected in two aspects. On the one hand, the original old samples can be obtained, and on the other hand, the model has certain storage resources. If conditions permit, retaining real samples can achieve a better incremental recognition effect. Therefore, the incremental learning method of conserving real exemplars is the focus of this article.

Since the model's recognition ability is derived from the effect of sample fitting, the selected samples' quality determines the recognition ability of the new model for the old samples. In order to pick high-quality samples, two exemplar selection methods are commonly used in incremental learning: exemplar selection based on herding [35] and random selection [34]. The former method picks the more representative samples that is closest to the class center in the feature space. However, these standard samples show a high similarity and a lack of diversity because of their short distances from the center, which is adverse to the model's training. Considering this problem, random selection can realize the random coverage of the whole class, which ensures the diversity of selected exemplars to a certain extent. However, the quality of picked samples may be different due to the instability of random selection.

Due to the SAR imaging mechanism, SAR images are more difficult to distinguish than their optical images, and images with small azimuth differences show obvious similarities. If representative samples can be selected according to the similarity between SAR samples, the size of SAR samples can be further reduced. Thus, to be aimed at the problems of the previous exemplar selection methods and SAR imaging mechanism, an exemplar selection method based on density is proposed from the perspective of maximum information coverage of the entire class. Different from the existing methods, density coverage-based exemplar selection (DCBES) considers the problem of exemplar selection as a set covering problem (SCP), which uses the greedy algorithm to gradually select representative samples to achieve the maximum coverage of the sample features. The experiment on the moving and stationary target acquisition and recognition (MSTAR) dataset proves that the exemplar selection method is more effective than the current mainstream method and achieves the most advanced effect. Fig. 1 shows the framework of incremental learning in this article. The exemplar selection part is DCBES algorithm proposed in this article.

II. Related Work

DCBES is an exemplar selection method used in incremental learning, so the related work mainly contains two parts: incremental learning and exemplar selection.

A. Incremental Learning

It is a method that can continuously learn new knowledge from the data stream without forgetting the old knowledge. The research fields of incremental learning can be divided into task incremental learning, class-incremental learning, and domain incremental learning [43].

Task incremental learning means that the data arriving at different times belong to different tasks, and the data of the same task can all arrive in one batch [23], [32], [37], [49]. Class-incremental learning refers to the data arriving at different times belonging to different classes of the same task, which can increase the classes of the output [17], [35], [53]. Domain incremental learning means that the data arriving at different times belong to the same class of the same task, but the data arrive in batches and the domain changes [39]. Since the mutual interference between the data before and after class-incremental learning is the largest, the difficulty is also the highest. This article focuses on class-incremental learning.

Class-incremental learning methods can be divided into approaches that are based on exemplars and not based on exemplars. The former can retain a handful of old class

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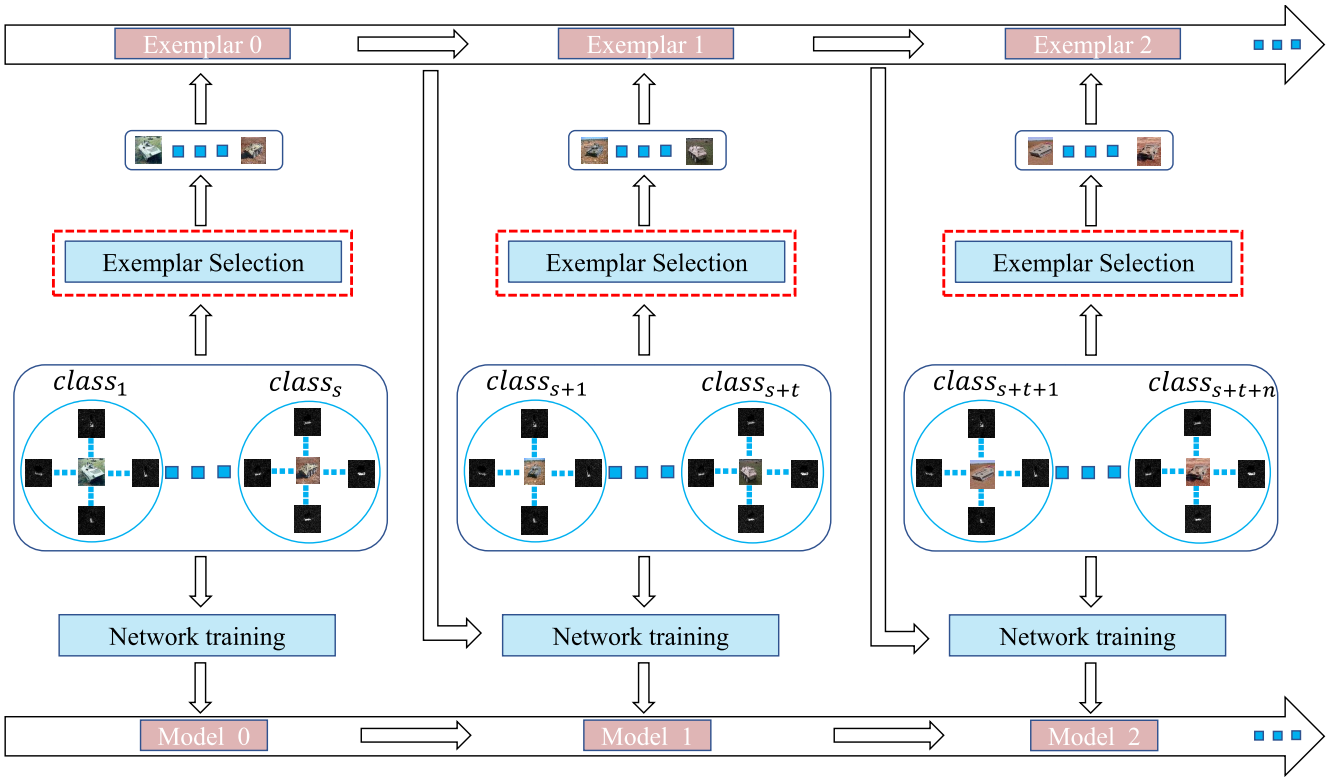


Fig. 1. Incremental learning scenario. Model 0 that can classify s classes is created by training the initial s classes. After training, the same number of exemplars is stored by DCBES for each class. When t new classes arrive, for the new classes, t new output nodes will be generated in the network to adapt to these correspondence classes. The s old exemplars will be trained together with the t new class samples. After that, model 0 is extended to model 1, and the same number of exemplars is stored for the new t classes.

samples or create old class samples through the generation model and then train them with the arriving new class samples. The effect is better than the incremental learning method without saving samples due to the preserved exemplars.

The first exemplar-based incremental learning method was proposed in [35], which uses the herding-based exemplar selection method and utilizes the nearest-mean-of-exemplars criterion for classification. This approach picks the samples closest to the class center because the author believes that those samples can best represent the whole class. After that, Castro et al. [9] and Wu et al. [45] and other methods also adopted the same exemplar selection method. Since random exemplar selection can ensure the diversity of selected samples to a certain extent and realize the random coverage of classes, which is also applied to incremental learning [34]. Considering the confidentiality of data and the effective utilization of storage resources, methods, such as [13], [24], [50], and [12], use GANs to generate old class samples without keeping them.

Incremental learning methods that are not based on examples are mainly divided into methods based on regularization and parameter isolation. The former mainly consolidates the previous knowledge when learning new data by adding extra regularization terms to the loss function. Li and Hoiem [27] proposed a method called learning without forgetting (LWF), which used the previous model output as the soft label of the previous task to reduce forgetting. Kirkpatrick et al. [23] developed an algorithm named elastic weight consolidation (EWC), and the parameters that were important to the old task should be punished so that they cannot change too fast.

The parameter isolation method mainly adds new structural branches for new tasks and freezes the old branch parameters while updating the branch parameters for new tasks. Rusu et al. [36] put forward a progressive neural network, which can utilize prior knowledge by laterally connecting to previously learned features. Xu and Zhu [48] raised reinforced continual learning (RCL), which searched for the best neural architecture for each upcoming task through a reinforcement learning strategy.

Due to the rapid development of incremental learning technology, it has been widely studied not only in the field of classification but also in the field of detection and segmentation. In the field of incremental detection, Joseph et al. [22] introduced open-world object detection, which can detect unknown classes and perform incremental detection step by step. A new remote sensing incremental object detection architecture was presented in [11], which was based on the feature pyramid and knowledge distillation. In the field of incremental segmentation, Cermelli et al. [10] proposed a new initialization strategy of objective function and classifier to effectively model the semantic changes of the background. Tasar et al. [42] applied incremental detection to large-scale remote sensing data and raised a method that enables neural networks to learn new classes while retaining the dense labeling ability of previously trained classes. Shan et al. [38] presented the deep feature distillation (DFD) module and the

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label mixed (LM) module for semantic segmentation in aerial imagery.

B. Exemplars Selection

For many datasets, there are correlations between many samples that produce a lot of redundant samples. The aim of exemplar selection is to choose key samples in different classes of the dataset and use a few samples to represent the whole class to the utmost extent, so as to reduce the size of the dataset.

In general, exemplar selection is based on model or training data distribution. The former is mainly based on some output indexes of the classification model, such as classification confidence, and samples with high classification confidence [14] will be picked first. The method based on training data's distribution chooses the exemplars according to the location of the samples in the feature space. Dang et al. [16] raised an exemplar selection method that mainly selected those samples located at the class boundary. Amrani et al. [3] proposed an efficient feature selection method based on the BoVW paradigm [40]. At present, the mainstream selection ways for incremental learning are two means based on data distribution: one is based on herding [35] and the other is random selection [34]. The exemplar selection based on herding picks the most standard samples that is closest to the class center in the feature space. However, these samples show a high similarity and a lack of diversity because of their short distances from the center. Random selection can realize the random coverage of the whole class, which ensures the diversity of selected exemplars to a certain extent. However, the quality of picked samples may be different due to the instability of random selection. This article focuses on the exemplar selection method based on the distribution of training data and considers the diversity and stability of the selected samples.

III. Exemplars Selection Based on Density Coverage for Class-Incremental Learning

This section will be described from two aspects: first the exemplar selection based on density coverage and then the incremental training.

A. Exemplars Selection Based on Density Coverage

In order to better understand the proposed algorithm, the theories of algorithm are introduced first. The proposed algorithm is mainly based on two theories, the metric learning theory and the set covering theory. First, introduce the metric learning theory and then the set covering theory.

Metric learning, also known as similarity learning, is a method of spatial mapping that can learn a feature (embedding) space. In this space, all data are converted into a feature vector, the distance between the feature vectors of similar samples is small, and the distance between the feature vectors of dissimilar samples is large, so as to distinguish the data. According to the metric learning theory [47], measuring the similarity between two samples $\boldsymbol{x}$ and $\boldsymbol{y}$ can learn a metric function

$$d(\boldsymbol{x}, \boldsymbol{y}) = d_A(\boldsymbol{x}, \boldsymbol{y}) = \|\boldsymbol{x} - \boldsymbol{y}\|_A = \sqrt{(\boldsymbol{x} - \boldsymbol{y})^T A(\boldsymbol{x} - \boldsymbol{y})}. \quad (1)$$

When A is positive semidefinite, the equation satisfies nonnegativeness and the triangle inequality. Obviously, given $A = I$, the Euclidean distance satisfies the above formula. Set pairs of points $(\boldsymbol{x}_i, \boldsymbol{x}_j) \epsilon S$ and S as a collection of similarity pairs. $(\boldsymbol{x}_i, \boldsymbol{x}_j) \epsilon D$, where D is the set of dissimilar point pairs. Xing et al. [47] treated metric learning as a convex optimization problem

$$\begin{aligned} &\min_A \sum_{(x_i, x_j) \epsilon S} \left\| x_i - x_j \right\|_A^2 \\ &\text{s.t.} \sum_{(x_i, x_j) \epsilon D} \left\| x_i - x_j \right\|_A \geq 1 \\ &\quad A \succeq 0. \end{aligned} \quad (2)$$

The Newton–Raphson method was used to prove in [47], and we will not repeat it here.

Due to the fact that the proposed exemplar selection method DCBES is used for selecting exemplars of a certain class and applied in the scenario of incremental learning, it is necessary to briefly describe the training status of each class of sample for incremental learning before proving the proposed method. This will help understand the proposed method.

In the process of incremental learning, when a sample passes through the fully connected (fc) layer of the network, it is mapped to an N-dimensional logit space, where N is the number of training classes. The network sets the corresponding anchored class centers (CACs) for samples of different classes, and each class corresponds to a CAC. The CAC c_i is equivalent to a scaled base vector e_i or a standard one-hot vector

$$\boldsymbol{C} = (\boldsymbol{c}_1, \ldots, \boldsymbol{c}_N) = (\alpha \cdot \boldsymbol{e}_1, \ldots, \alpha \cdot \boldsymbol{e}_N) \quad (3)$$

$$\boldsymbol{e}_1 = (1, 0, \ldots, 0)^T, \boldsymbol{e}_N = (0, \ldots, 0, 1)^T. \quad (4)$$

In the training process, samples of different classes will gather near the corresponding CACs to form clusters. The formation of clusters is formed by the CAC loss set by training, which includes two parts: the anchored loss $\mathcal{L}_A(\boldsymbol{x}, y)$ and the discrimination loss $\mathcal{L}_D(\boldsymbol{x}, y)$. The definitions of the two losses are shown as follows:

$$\mathcal{L}_A(\boldsymbol{x}, y) = d_y = \left\| f(\boldsymbol{x}) - \boldsymbol{c}_y \right\|_2 \quad (5)$$

$$\mathcal{L}_D(\boldsymbol{x}, y) = \log\left(1 + \sum_{j \neq y}^{N} e^{d_y - d_j}\right). \quad (6)$$

For $\mathcal{L}_A(\boldsymbol{x}, y)$, $f(x)$ represents the logits through the network fc layer. Through the constraint of $\mathcal{L}_A(\boldsymbol{x}, y)$, samples of all classes will gather near the corresponding CACs. For $\mathcal{L}_D(\boldsymbol{x}, y)$, d_y and d_j, respectively, represent the distance from $f(x)$ to the corresponding CAC and other CACs. Through the constraint of $\mathcal{L}_D(\boldsymbol{x}, y)$, samples of different classes will move away from each other. Therefore, the definition of the total loss $\mathcal{L}_{\text{CAC}}(\boldsymbol{x}, y)$ for training is given as follows:

$$\mathcal{L}_{\text{CAC}}(\boldsymbol{x}, y) = \mathcal{L}_D(\boldsymbol{x}, y) + \lambda \cdot \mathcal{L}_A(\boldsymbol{x}, y) \quad (7)$$

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where λ is a hyperparameter that balances these two individual loss terms. Through the constraint of $\mathcal{L}_{\mathrm{CAC}}(\boldsymbol{x}, y)$, samples of all classes will gather near the corresponding CACs to form clusters, that is to say, the proposed exemplar selection method is to select features for each cluster and save the samples corresponding to the features as exemplars. In the following, we prove that our loss function $\mathcal{L}_{\mathrm{CAC}}(\boldsymbol{x}, y)$ conforms to metric learning theory.

When $A = I$, at the same time

$$\begin{aligned} d(f(\boldsymbol{x}_i), f(\boldsymbol{x}_j)) &= \left\| f(\boldsymbol{x}_i) - f(\boldsymbol{x}_j) \right\|_2 \\ &\le \left\| f(\boldsymbol{x}_i) - \boldsymbol{c}_y \right\|_2 + \left\| f(\boldsymbol{x}_j) - \boldsymbol{c}_y \right\|_2 \\ &= d(f(\boldsymbol{x}_i), c_y) + d(f(\boldsymbol{x}_j), c_y) \\ &= \mathcal{L}_A(x_i, y) + \mathcal{L}_A(x_j, y). \end{aligned} \tag{8}$$

Measure the sum of the distances between $f(\boldsymbol{x}_i)$ and $f(\boldsymbol{x}_j)$ to c_y is equivalent to measuring the upper bound of the distance between $f(\boldsymbol{x}_i)$ and $f(\boldsymbol{x}_j)$. Thus, $\mathcal{L}_A(\boldsymbol{x}, \boldsymbol{y})$ satisfies the metric learning theory.

Similarly, for $\mathcal{L}_D(\boldsymbol{x}, y)$

$$\begin{aligned} \mathcal{L}_D(\boldsymbol{x}, y) &= \log\left(1 + \sum_{j \neq y}^{N} e^{d_y - d_j}\right) \\ &= \log\left(1 + \sum_{j \neq y}^{N} e^{\left\| f(\boldsymbol{x}) - \boldsymbol{c}_y \right\|_2 - \left\| f(\boldsymbol{x}) - \boldsymbol{c}_j \right\|_2}\right). \end{aligned} \tag{9}$$

Combining (8) and (9), a discriminatory loss $\mathcal{L}_D(\boldsymbol{x}, y)$ also satisfies the theory of metric learning. Therefore, the CAC loss function $\mathcal{L}_{\mathrm{CAC}}(\boldsymbol{x}, y)$ that is used in incremental learning conforms to the theory of metric learning.

Since the incremental loss function conforms to the metric learning theory, it is feasible for the proposed algorithm to use the neighborhood of samples in the metric space to divide density subsets. Next, we will prove that the proposed algorithm DCBES conforms to the set covering theory.

The SCP is a nondeterministic polynomial (NP)-complete problem, and there is no polynomial time algorithm for this problem. The mathematical description of the set coverage problem is given as follows.

$< X, F >$ is a finite set that is composed of a subset family F of X. The subfamily F covers a finite set, so every element in X belongs to at least one subset of F, $X = \cup_{S \epsilon F} S$. For a subset $C \subseteq F$, if a subset of X in C covers X, $X = \cup_{S \epsilon C} S$, C is said to cover X. The set coverage problem is to find the smallest set C that covers X in F

$$\begin{aligned} &\min \mid C \mid \\ &\text{s.t. } X = \bigcup_{S \epsilon C} S. \end{aligned} \tag{10}$$

However, unlike the traditional SCP, this article does not require finding the minimum number of sets C^* that can completely cover X but requires that the maximum coverage of X can be sought under the condition that num (C^*) is equal to a fixed value m

$$\begin{aligned} &\max \mid \bigcup_{S \epsilon C} S \mid \\ &\text{s.t. } \mid C \mid = m. \end{aligned} \tag{11}$$

Next, we will prove that our exemplar selection problem conforms to the maximum set covering theory.

According to the metric learning theory, for a certain class, all samples of that class are mapped to the metric space. According to (7) and (8), the neighborhood of each feature is also determined. The set of features in the entire class can be defined as the complete set X, and the neighborhood N_y^j of each feature can be defined as a subset of X, so $X = \cup_{S^j = N_y^j} S^j$.

Since the purpose of exemplar selection is to use some of the samples selected in this class to represent the entire class to the greatest extent, considering the impact of storage resources, the number of samples selected should be limited and small.

Based on the metric learning theory, samples with high similarity are close in distance and contain less information, while samples with low similarity are far away and contain more information. Therefore, the purpose of our algorithm DCBES is to select a fixed number of samples to contain more information of the class. From the set covering theory, it is to seek the maximum coverage of the complete set X under the condition that the number of subsets C is fixed. SCP is generally solved by a greedy algorithm. Assuming that n subsets $S^1, \ldots, S^n$ have been selected and the union of the sets is expressed as: $\hat{X} = \bigcup_{S^i = N_y^i}^{i=1,\ldots,n} S^i$, where X_{total} is the total density subset, then the mathematical description of selecting the next set is given as follows:

$$S^j = \max\left\{ \mid S^j \bigcup \hat{X} \mid\mid S^j \epsilon X_{\mathrm{total}} - S^1 - \cdots - S^n \right\}. \tag{12}$$

Therefore, the proposed algorithm DCBES conforms to the set covering theory. In summary, the proposed algorithm conforms to the metric learning theory and the maximum set covering theory.

In the metric space, the number of other sample features in the neighborhood of one certain sample determines the density of this specific sample. According to the theory of metric learning, the larger the density, the more samples similar to this certain sample and vice versa. Apparently, this specific sample can be regarded as the representative exemplar of this neighborhood space. In the process of exemplar selection, based on the maximum set covering theory, the picked samples can stand for the whole class if the set of neighborhood space of these samples can achieve the maximum coverage of the whole class.

Define the sample sets $X = (X_1, X_2, \ldots, X_s)$, where X_y contains all images from class y. For all the images of X_y, the corresponding feature set $F_y = (f_y^1, f_y^2, \ldots, f_y^n)$ will be obtained after passing through the network, where n represents the number of samples in class y. The raised exemplar selection method picks the corresponding samples according to the distribution of feature set F_y in the feature space. The density of feature points f_y^j is defined as the subsample set N_y^j whose distance from f_y^j is not greater than ε in the feature set F_y. ε represents the neighborhood radius of f_y^j, that is, $N_y^j = \{f_y^i \epsilon F_y | \mathrm{distance}(f_y^i, f_y^j) \le \varepsilon\}$. The set of features in the entire class can be defined as the complete set X, and the neighborhood N_y^j of each feature as a subset of

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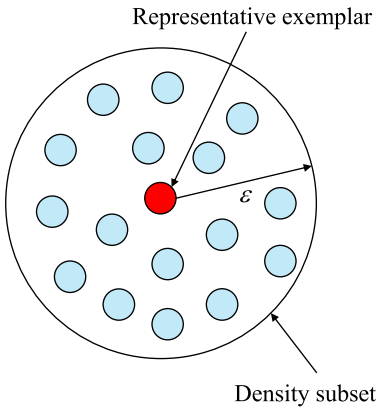


Fig. 2. Schematic of the density subset. The red circle represents the representative exemplar, the blue circles represent the samples in the neighborhood, and ε represents the neighborhood radius.

X. Thus, $X = \cup_{S^j = N_y^j} S^j$. Fig. 2 shows the schematic of the density subset N_y^j.

Therefore, as long as the neighborhood ε is determined, the problem of exemplar selection is transformed into the problem of maximum set covering, and only the exemplars corresponding to the set need to be selected. Next, the determination of neighborhood ε will be described in detail.

Inspired by the density-based spatial clustering of applications with noise (DBSCAN) [18] algorithm, the following definitions can be made:

$$t = 100/m \tag{13}$$

where t represents that the nearest point around each point accounts for t percent of the total points and m represents the number of selected exemplars. Unlike DBSCAN, t can be determined directly from m in our method, while t in DBSCAN needs to be set by human experience. The reason for this setting is that this formula indicates that after m exemplar selection, the density area of the selected exemplars can achieve the overall coverage of the whole class in an average sense. Then, the neighborhood radius ε can be defined as

$$\varepsilon = \text{choseepsilon}(\text{Distance}, t) \tag{14}$$

where Distance represents the distance matrix between all feature points in this class, and the size is $n \times n$. For the implementation of the chooseepsilon method of selecting ε, please refer to DBSCAN [18] for details.

In the process of exemplar selection, the values of m and ε are set, and the density subset N_y^j is determined. In the process of exemplar selection, a greedy algorithm is used to select those exemplars with the largest union with the selected sample set in the remaining sample set, as described in (12).

Algorithm 1 exhibits the details of algorithm flow. Since our algorithm, DCBES, is based on the set covering theory and the SCP is an NP problem, the limitation of the algorithm is mainly reflected in the high computational complexity of the algorithm. Therefore, the algorithm is mainly applicable to incremental learning of small- and medium-sized datasets, not large datasets.

Algorithm 1 DCBES

Require: $F_y = (f_y^1, f_y^2, \ldots, f_y^n)$ //features from class y
Require: m // The number of selected exemplars,
Require: t // the nearest point around each point accounts for t percent of the total points
for $i = 1, \ldots, n$
  for $j = 1, \ldots, n$
    if $i \geq j$
      $distance[i][j] \longleftarrow distance[j][i]$
    else
      $distance[i][j] \longleftarrow \left\| F_y^i - F_y^j \right\|$
$\varepsilon \longleftarrow choseepsilon(distance, t)$
for $j = 1, \ldots, len(F_y)$
  for $k = 1, \ldots, len(F_y)$
    $N_y^j \longleftarrow \left\{ f_y^j \epsilon F_y | distance(f_y^j, f_y^k) \leq \varepsilon \right\}$
$N_y = \left\{ N_y^1, N_y^2, \ldots, N_y^{len(F_y)} \right\}$ //Total density subset
$M_y = \phi$ //Initialize the selected density subset
for $i = 1, \ldots, m$
  $S^i = max\left\{ | S^i \cup M_y | | S^i \epsilon N_y - M_y \right\}$
  $M_y = \left\{ S^1, S^2, \ldots, S^i \right\}$
  $P_y^i \longleftarrow X_y^i$
Ensure: $P_y = \left\{ P_y^1, P_y^2, \ldots, P_y^m \right\}$

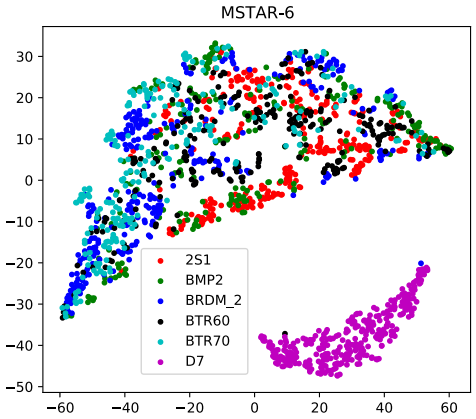


Fig. 3. Result of the t-SNE dimensionality reduction visualization of six classes (2S1, BMP2, BRDM2, BTR60, BTR70, and D7). These five classes' features (2S1, BMP2, BRDM2, BTR60, and BTR70) are extremely similar and the features of the same class are relatively diverged.

B. Incremental Learning

The incremental training method determines not only the use of the new and the old class samples but also the update of the old and new models. Next, the incremental training method we use will be introduced in detail.

Considering that the SAR target is characterized by large intraclass distance and small interclass distance, the distributions of new class and old class are prone to aliasing, which leads to the error discrimination of the model. To prove that, t-distributed stochastic neighbor embedding (t-SNE) [44] visualization method is used for low-dimensional visualization for six categories of samples in the MSTAR dataset, as shown in Fig. 3. t-SNE is an embedding model that can map data in high-dimensional space to low-dimensional space and retain the local characteristics of the dataset. After the t-SNE

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transformation, if it is separable in the low-dimensional space, the data are separable; otherwise, the data are not separable. As can be seen in Fig. 3, the sample characteristics of five classes, 2S1, BMP2, BRDM2, BTR60, and BTR70, are very similar and the distances between classes are small. At the same time, the feature distribution of each class is relatively scattered with a larger intraclass distance. According to the literature [46], if the classes in the incremental process are more similar, then the incremental accuracy will be reduced significantly.

Algorithm 2 Training Step

Require: $\mathcal{X} = (X_{s+1}, \ldots, X_{s+t}), \mathcal{P} = (P_1, \ldots, P_s)$ // new classes data, old exemplar sets
Require: $\Theta^s, \hat{\Theta}^s, A_s$ // current model, frozen current model, current anchored class centers
Ensure: Θ^{s+t} // model trained on t new classes
$\Theta^{s+t} \leftarrow \Theta^s$ // add output nodes for new classes
$A_{s+t} \leftarrow A_s$ // add new anchored class centers for new classes
for $(x, y) \epsilon\ \mathcal{X}$ **do**
  update for mini-batch data in $\mathcal{X}$
    $o = \Theta^{s+t}(x)$ // $o = \{o_{old}, o_{new}\}$.
    compute classification loss $\mathcal{L}_{\mathcal{X}}^{CAC}$ with o (Eq. 7)
    softmax over old class logits $\sigma(o_{old})$
    compute distillation loss $\mathcal{L}_{\mathcal{X}}^{KD}$ (Eq. 15)
    load a mini-batch from exemplars set$(x', y') \epsilon\ \mathcal{P}$
    $o' = \Theta^{s+t}(x')$ // $o' = \{o'_{old}, o'_{new}\}$
    compute classification loss $\mathcal{L}_{\mathcal{P}}^{CAC}$ with o' (Eq. 7)
    softmax over old class logits $\sigma(o'_{old})$
    compute distillation loss $\mathcal{L}_{\mathcal{P}}^{KD}$ (Eq. 16)
    $\mathcal{L} = (\mathcal{L}_{\mathcal{X}}^{CAC} + \mathcal{L}_{\mathcal{P}}^{CAC}) + \xi * (\mathcal{L}_{\mathcal{X}}^{KD} + \mathcal{L}_{\mathcal{P}}^{KD})$
end

To solve this problem, in this article, incremental class anchor clustering (ICAC) [25], which is based on the CAC, is adopted. ICAC selects exemplars by random selection. The main contribution of this article is to propose a new exemplar selection method for ICAC.

Initially, a model is trained with the known s classes. For the samples of these s classes, the model will map them to the s-dimensional feature space at the last layer. An s-dimensional coordinate system is established according to the s-dimensional feature space and there is a CAC point on each coordinate axis corresponding to each class.

According to (3)–(7), in the training phase, all samples of each class will gather around the corresponding CAC. For samples of different classes, the distance between classes is large because the class anchor points are located on different coordinate axes. For a certain class, the distance within the class is small since the distance between the samples gathered around the anchor point is small. Therefore, each class can be effectively distinguished through such training.

When t new class samples arrive, on the one hand, m exemplars will be picked for each old class by the exemplar selection method DCBES. The chosen exemplars will be trained together with the new class samples. On the other hand, the dimension of the feature space will be increased. The added dimension is consistent with the number of new classes. At the same time, t CACs will be set on the coordinate axis of the new dimension, and each new class corresponds to a new CAC. In the training stage, the original old samples will be gathered near the corresponding old anchor points, and the new class samples will be gathered near the corresponding new anchor points. In this way, the differences between classes can be increased and the differences within classes can be decreased, so as to effectively prevent the old and new class samples from overlapping in distribution.

In terms of the updating modes of the old and new models, on the one hand, the network architecture will be updated. When t new classes arrive, t new output nodes are added adaptively at the last layer of the network, which will be initialized and connected in parallel with the previous s nodes. Therefore, the new model will contain $s + t$ output nodes. When the new class sample x passes through the network, the output $o = \{o_{\text{old}}, o_{\text{new}}\}$ will be get, where o_{old} represents the output of the new class sample about the old node and o_{new} indicates the output of the new class sample about the new node. When the old exemplar x' passes through the network, the output $o' = \{o'_{\text{old}}, o'_{\text{new}}\}$ will be get, where o'_{old} indicates the output of the old exemplar about the old node and o'_{new} represents the output of the old exemplar about the new node.

On the other hand, knowledge distillation is utilized to constrain and update the model parameters. The method of parameter updating is given as follows:

$$\mathcal{L}_{\mathcal{X}}^{\text{KD}} = D_{\text{KL}}\left(\sigma(\hat{\Theta}^s(x)) \| \sigma(o_{\text{old}})\right) \tag{15}$$

$$\mathcal{L}_{\mathcal{P}}^{\text{KD}} = D_{\text{KL}}\left(\sigma(\hat{\Theta}^s(x')) \| \sigma\left(o'_{\text{old}}\right)\right) \tag{16}$$

where D_{KL} represents the Kullback–Leibler (KL) divergence, σ represents the softmax over class logits, and $\hat{\Theta}$ represents the old model.

In order to solve the imbalance between the new and old class samples, ICAC uses the strategy of separation learning (SL), and SL computes two separate losses for new class set X and exemplar set P mini-batches and adds the two losses to make a gradient descent

$$\mathcal{L}_{\mathcal{X}} = \mathcal{L}_{\mathcal{X}}^{\text{CAC}} + \xi * \mathcal{L}_{\mathcal{X}}^{\text{KD}} \tag{17}$$

$$\mathcal{L}_{\mathcal{P}} = \mathcal{L}_{\mathcal{P}}^{\text{CAC}} + \xi * \mathcal{L}_{\mathcal{P}}^{\text{KD}}. \tag{18}$$

The adaptive weighting function ξ between the classification loss and the distillation loss is defined as follows:

$$\xi = \xi_{\text{base}} \left(\frac{C_n + C_0}{C_n}\right)^{2/3} \tag{19}$$

where C_n denotes the number of new classes and C_0 denotes the number of old classes. ξ_{base} is 5.

To better understand the training process, Fig. 4 shows the schematic of the incremental process from two to three classes.

In the test phase, the algorithm will calculate the class prototype for the exemplars corresponding to each class, and the output test samples will be classified using the nearest neighbor criterion. The specific training and testing details are shown in Algorithms 2 and 3, respectively.

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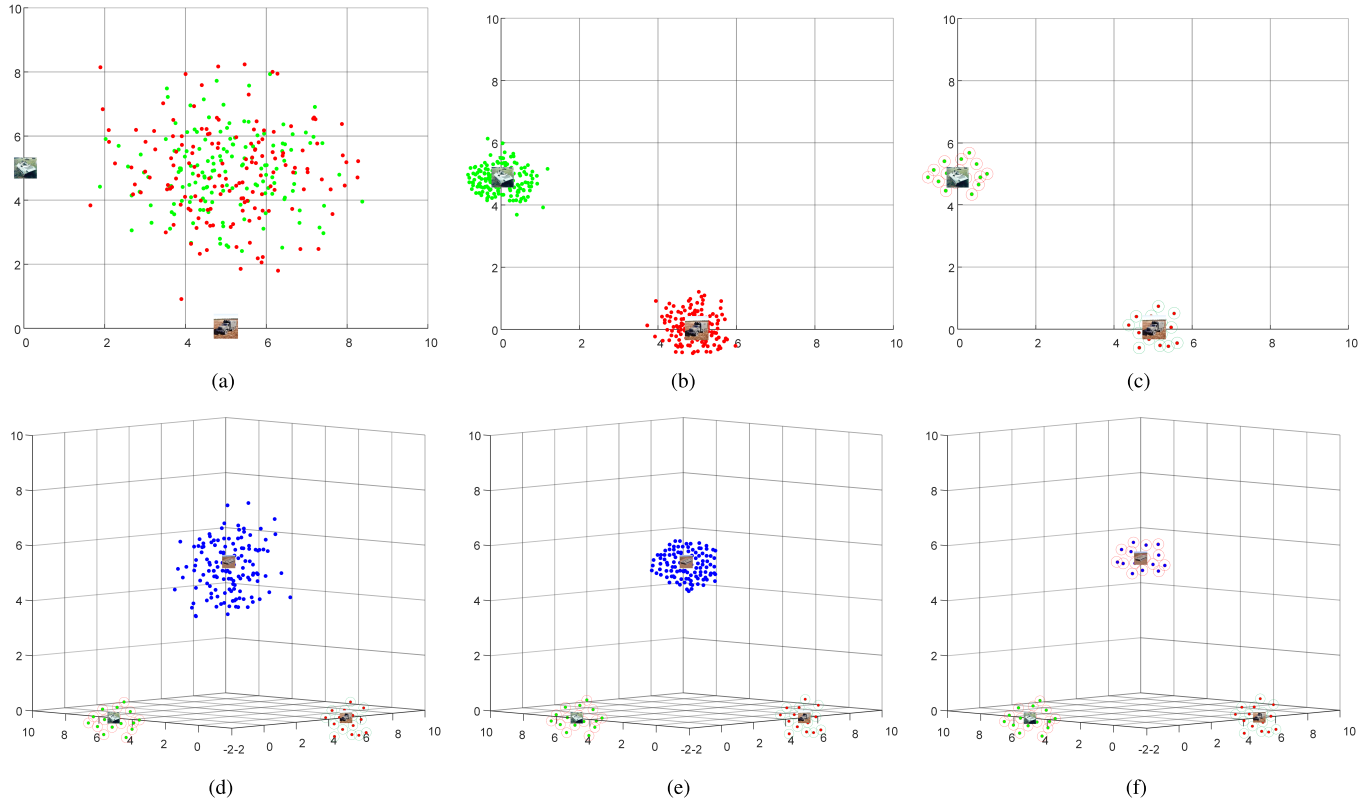


Fig. 4. Schematic of the incremental process from two to three classes. (a) Red and green dots represent the corresponding features of the two class samples, and the position of the image on the coordinate axis represents the position of the CAC. (b) Features of different classes are clustered around the corresponding CACs after training. (c) Solid dots represent m samples selected as exemplars by DCBES for each class. (d) Blue dots represent new class features. When a new class arrives, the feature space is increased from 2-D to 3-D, and a new CAC will be generated in the network to adapt to the new class. (e) Features of new class are all clustered around the new CAC. (f) Select m samples as exemplars for the new class.

Algorithm 3 Test Step

Require: x // image to be classified
Require: $\mathcal{P} = (P_1, \ldots, P_{s+t})$ // class exemplar sets
Require: $\Theta^{s+t} : \mathcal{X} \longrightarrow \mathbb{R}^{s+t}$ // feature map
1: **for** $y = 1, \ldots, s+t$ **do**
2: $\mu_y \longleftarrow \frac{1}{|P_y|} \sum_{p \epsilon P_y} \Theta^{s+t}(p)$ // mean-of-exemplars
3: **end for**
4: $y^* \longleftarrow \arg\min_{y=1,\cdots,s+t} \left\| \Theta^{s+t}(x) - \mu_y \right\|$ // nearest prototype
Ensure: class label y^*

IV. Experimental Evaluation

A. Experimental Setup

1) Dataset: Experiments on the MSTAR dataset were performed. The MSTAR dataset is organized under a Defense Advanced Research Projects Agency (DARPA) program. The collecting sensor is a high-resolution spotlight SAR with a resolution of 0.3 × 0.3 m, which works in the X-band, and the used polarization mode is HH polarization mode. The collected data are preprocessed and slice images containing various targets with a pixel size of 128 × 128 are extracted. The MSTAR dataset contains ten classes of target samples, namely, 2S1 (cannon), BMP2 (tank), BRDM2 (truck), BTR60 (armored car), BTR70 (armored car), D7 (bulldozer), T62 (tank), T72 (tank), ZIL131 (truck), and ZSU23/4 (cannon). These targets are captured at two different depression angles (15° and 17°) with 190–300 different aspect versions. The details of the MSTAR dataset are shown in Table I. The number after the target type represents the type code. In the experiments, the images with depression angles of 17° and 15° are applied as the training set and the test set severally, respectively. In the data preprocessing stage, all images are resized to 64 × 64 but without any other preprocessing. For the MSTAR dataset, four random classes are initial training classes and the remaining six classes are incremental training classes. All experiments are the average performance under five different splits. Table II shows five different ways of dividing the initial training classes and incremental classes.

2) Comparison With Current Methods: In order to effectively combine with our incremental learning method, all the comparison methods are based on data distribution. Therefore, four algorithms based on data distribution are compared, i.e., pick exemplars based on herding [35], edge [16], randomly pick [34], and class boundary exemplar selection-based incremental learning (CBesIL) [17]. Rebuffi et al. [35] used the herding-based exemplar selection method, which continuously selects the samples close to the class center as exemplars. Dang et al. [16] selected the samples closest to the boundary as exemplars. Mittal et al. [34] used random seeds to randomly select exemplars. Since Dang et al. [17] selected complete boundary samples as exemplars and the

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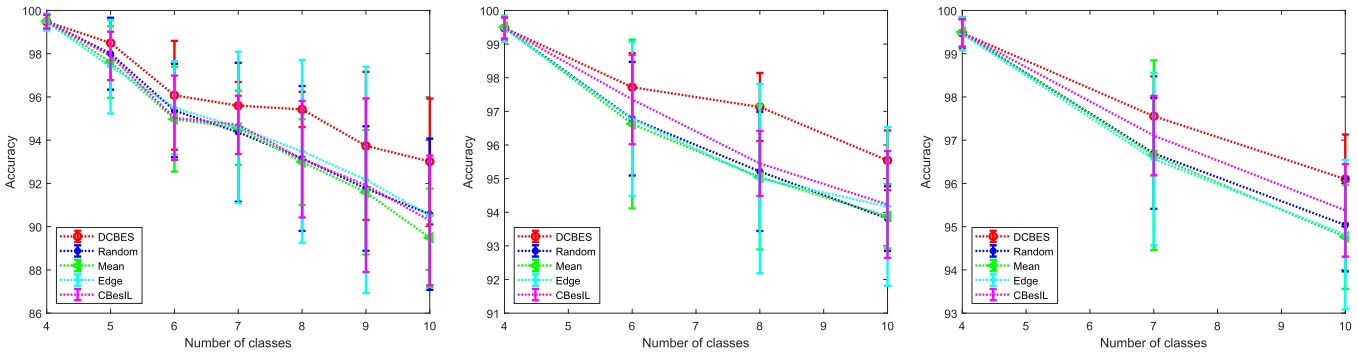


Fig. 5. Multiclass accuracy (averages and standard deviations over five different splits) on the MSTAR dataset under the condition of different incremental steps T. (a) Add one class per incremental step ($T = 1$). (b) Add two classes per incremental step ($T = 2$). (c) Add three classes per incremental step ($T = 3$).

TABLE I
SUMMARY OF THE USED MSTAR DATABASE

Target Type	Training Set (17°)	Testing set (15°)
2S1 (0)	299	274
BMP2 (1)	233	195
BRDM2 (2)	298	274
BTR60 (3)	256	195
BTR70 (4)	233	196
D7 (5)	299	274
T62 (6)	299	273
T72 (7)	232	196
ZIL131 (8)	299	274
ZSU23/4 (9)	299	274

TABLE II
FIVE DIFFERENT SPLITS OF TRAINING CLASSES

Initial training classes	Incremental training classes
(9,5,2,4)	(7,1,0,8,6,3)
(8,1,7,0)	(6,5,2,4,3,9)
(3,5,6,1)	(4,7,8,9,0,2)
(3,9,0,5)	(4,2,1,7,6,8)
(7,1,8,5)	(0,2,6,9,4,3)

comparison of each method is based on a fixed number of exemplars, we adopt uniform sampling of boundary samples from different angles to achieve the predetermined number of exemplars.

The following groups of experiments are set up. First, the class-incremental recognition performance of each method under different incremental steps is compared. The class-incremental step is set to be $T = 1, 2, 3$ since the different incremental steps will be used for incremental learning in practical applications. Then, the class-incremental recognition performance of each method under different saved exemplars m is considered. In this experiment, m is set to be 20, 30, and 40, which is caused by the possible limited storage resources in the practical application of incremental learning. At the same time, assume that only 10%–20% of the training data can be stored. Next, the class-incremental recognition performance of each method under different anchor magnitudes $\alpha = 1, 5, 20$ is explored. Finally, the test data distribution of each method is visualized when the incremental training was up to seven classes and eight classes in order to observe a specific distribution of each class in the incremental process under different exemplar selection methods.

3) Evaluation Protocol: The standard metrics are reported to measure the quality of CIL. Accuracy [35] is computed as the average accuracy of all the classes that have already been learned.

4) Implementation Details: For the feature extractor network, the network proposed in [28] is adopted and the fc(·) layers are modified in the experiments. The models are trained with a batch size of 16 and an adaptive moment estimation (Adam) optimizer with a 0.00001 learning rate. The training epoch is 100 for each method, the anchor magnitudes $\alpha = 5$ for experiments 1 and 2, and the anchor loss weight $\lambda = 0.1$.

B. Class-Incremental Recognition Performance

1) Under Different Incremental Steps T: Fig. 5 shows the incremental recognition performance of different exemplar selection methods under different incremental steps ($T = 1, 2, 3$). Three phenomena are found in Fig. 5. First, our proposed algorithm DCBES achieves the best incremental recognition effect under different incremental steps T. Second, with the gradual increase of the incremental steps, the difference between each method becomes smaller. The main reason is that the gradual increase of the incremental steps represents the gradual increase of the number of new classes in each incremental process, so the proportion of new class samples in each training process gradually increases. Therefore, the incremental recognition effect of the model mainly depends on the new class samples. Third, as the incremental steps gradually increase, the forgetting speed of the model gradually slows down. Since each update of the model means the refitting of the classification boundary of the model, the more the fitting times, the faster the model forgets the old samples. Therefore,

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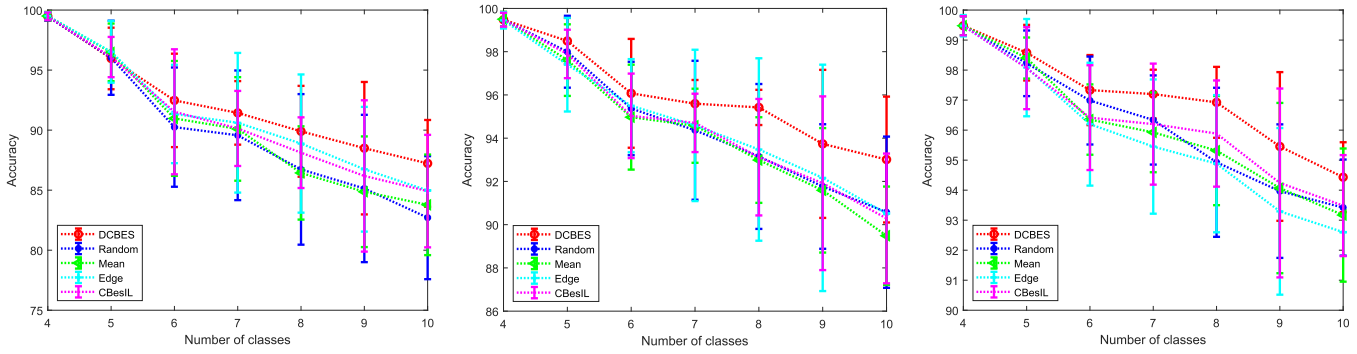


Fig. 6. Multiclass accuracy (averages and standard deviations over five different splits) on the MSTAR dataset under the condition of different exemplar set sizes m for each class. (a) Exemplar set sizes $m = 20$. (b) Exemplar set sizes $m = 30$. (c) Exemplar set sizes $m = 40$.

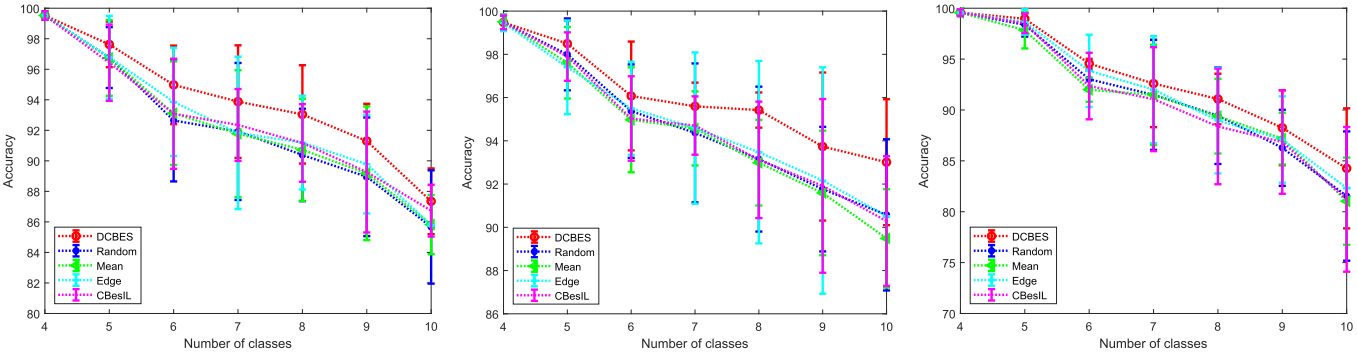


Fig. 7. Multiclass accuracy (averages and standard deviations over five different splits) on the MSTAR dataset under the condition of different anchor magnitudes α for each class. (a) Anchor magnitudes $\alpha = 1$. (b) Anchor magnitudes $\alpha = 5$. (c) Anchor magnitudes $\alpha = 20$.

when the number of incremental steps is fixed, increasing the incremental steps is conducive to slowing down the forgetting of the model. Of course, the setting of incremental steps should be made according to the requirements of the actual application scenario.

2) Under Different Preserved Exemplars m: Fig. 6 shows the performance of different exemplar selection methods when the number of preserved exemplars is ($m = 20, 30, 40$) and the incremental step is 1. Three phenomena are found in Fig. 6. First, the algorithm DCBES proposed in this article achieves the best effect in different numbers of exemplars m. Second, with the increase of m, the difference between different exemplar selection methods decreases (note the numerical change of the ordinate). The main reason is that with the increase of reserved sample m, the distribution of old class is better restored. Third, when the reserved sample m is smaller, the advantages of DCBES over other algorithms are more obvious. Since no matter how many exemplars are selected, the exemplars selected by DCBES can achieve the maximum coverage of the old class distribution. Therefore, when m is smaller, DCBES can pick more representative samples than other methods.

3) Under Different Anchor Magnitudes α: Fig. 7 shows the effects of different anchor magnitudes α, which represents the positions of the class anchor points on the axis. The anchor magnitude α will affect the aggregation of each class of sample during the training process, and it will affect the distribution of class samples. If the anchor magnitude α is too large, then the training of the network will be difficult to converge. If the magnitude α is too small, then the distance between different classes will be very small, resulting in wrong judgments between classes. It can be observed that when α is large, the difference between different methods is smaller. The main reason is that when α is too large, the distribution of classes is more dispersed, resulting in a greater distance between samples. Therefore, the effect of using Euclidean distance to measure sample similarity is not obvious.

4) Incremental Visualization Effect of Each Method Under Reserved Exemplars Number m = 30: Fig. 8 shows the visual results on the test set when each method incrementally learns to seven classes and eight classes. Dots of different colors represent different test classes, and color icons stand for the class order of increment from top to bottom. In Fig. 8(a)–(e), it can be seen that when the four exemplar selection methods, random, edge, mean, and CBesIL, are used for increment, the distribution of 2S1 and the new arriving class BRDM2 during the increment stage are mixed. The mixing of these two indicates that the old class is easy to be misclassified into the new class in the increment process, especially the old class samples close to the new class distribution.

However, from Fig. 8(e), we can see that the exemplar selection method based on DCBES can effectively avoid the distribution mixing between the new class BRDM2 and the

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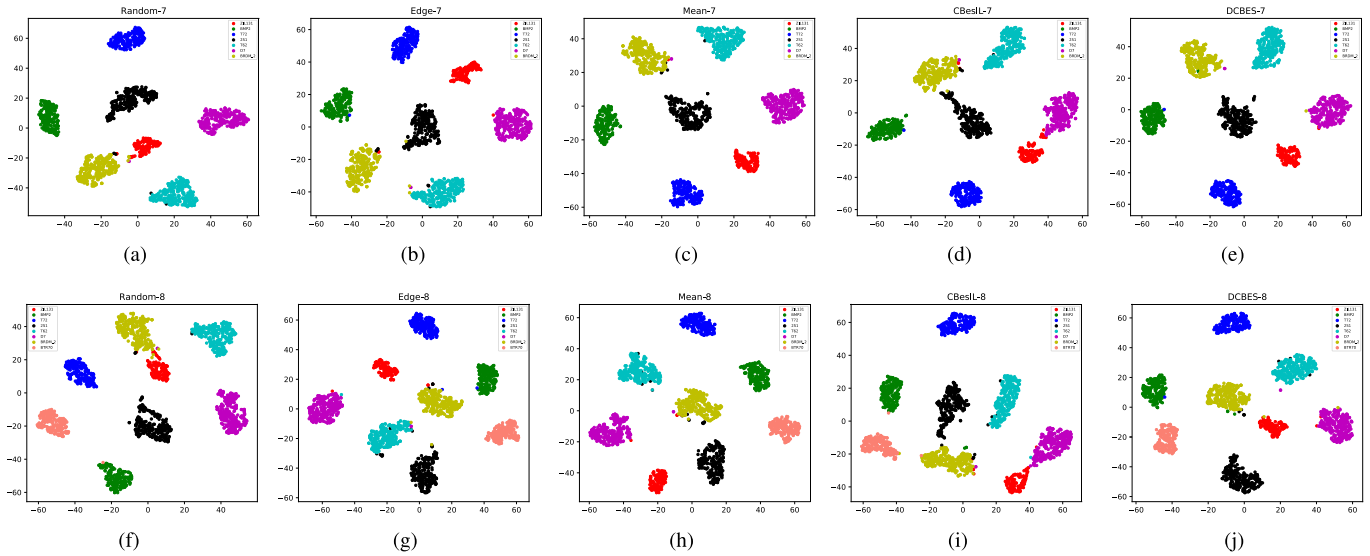


Fig. 8. Visual results on the test set when each method incrementally learns to seven classes and eight classes, the initial training class is 4. (a) Random-based incremental learning to class 7. (b) Edge-based incremental learning to class 7. (c) Mean-based incremental learning to class 7. (d) CBesIL-based incremental learning to class 7. (e) DCBES-based incremental learning to class 7. (f) Random-based incremental learning to class 8. (g) Edge-based incremental learning to class 8. (h) Mean-based incremental learning to class 8. (i) CBesIL-based incremental learning to class 8. (j) DCBES-based incremental learning to class 8.

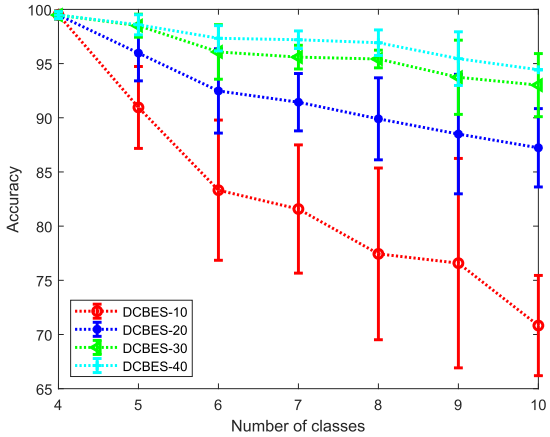


Fig. 9. Multiclass-incremental accuracy of DCBES under a different number of saved exemplars.

old classes. In Fig. 8(e)–(j), it can be seen that DCBES can effectively avoid the distribution aliasing between the new class and the old classes.

5) Additional Discussion on Different Preserved Exemplars m: Considering that in real-world application scenarios, there are some requirements for multiclass accuracy of incremental models, that is, to achieve a certain accuracy before practical application. Fig. 9 shows the multiclass-incremental accuracy of DCBES under different numbers of saved exemplars. From Fig. 9, it can be seen that in the process from four types of increment to ten types of increment, if the accuracy of the model is required to be no less than 70%, only ten exemplars need to be retained. If the accuracy is required to be no less than 90%, only 20 exemplars need to be saved. If the accuracy is required to be no less than 95%, only 30 exemplars need to be preserved. This experiment provides a reference for the number of samples that need to be kept in practical applications.

V. Conclusion

In this article, a new exemplar selection method DCBES has been proposed for incremental SAR ATR, which is based on the density of samples in the feature space. The purpose of DCBES is to pick those samples that can cover the class distribution to the maximum extent as exemplars. DCBES considers exemplar selection as an SCP and uses the greedy algorithm to choose representative samples. The experiment on the MSTAR dataset shows that the raised method has the best incremental recognition effect compared with the current mainstream exemplar selection method of incremental learning, whether under different incremental steps or under different exemplar numbers. The test visualization in the incremental process also further verifies that the proposed method can select the most representative examples, thus effectively avoiding the model bias to recognize old classes as new classes.

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Bin Li received the B.E. degree in communication engineering from the Chongqing University of Posts and Telecommunications, Chongqing, China, in 2019. He is currently pursuing the Ph.D. degree with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu, China.

His research interests include synthetic aperture radar (SAR) image processing and incremental learning.



Zongyong Cui (Member, IEEE) received the B.E. and Ph.D. degrees in signal and information processing from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2007 and 2015, respectively.

From 2013 to 2014, he was a Visiting Student with the Department of Electrical and Computer Engineering, National University of Singapore, Singapore. He is currently an Associate Professor with the School of Information and Communication Engineering, UESTC. His research interests include radar image processing, target recognition, and machine learning.

Dr. Cui is a reviewer of the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, the *IET Radar, Sonar and Navigation*, and *Journal of Electronic Imaging*.



Yuxuan Sun received the B.E. degree in communication engineering from Xidian University, Xi'an, China, in 2020, and the M.Eng. degree from the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu, China, in 2023.

His research direction is synthetic aperture radar (SAR) image recognition under imbalanced data.



Jianyu Yang (Member, IEEE) received the B.S. degree from the National University of Defense Technology, Changsha, China, in 1984, and the M.S. and Ph.D. degrees from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 1987 and 1991, respectively.

He is currently a Professor with UESTC. His research interests include synthetic aperture radar and statistical signal processing.

Dr. Yang serves as a Senior Editor for the *Chinese Journal of Radio Science* and the *Journal of Systems Engineering and Electronics*.



Zongjie Cao (Member, IEEE) received the B.E. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1999, and 2005, respectively.

From 2006 to 2008, he was a Post-Doctoral Researcher with the Communication and Information System Postdoctoral Center, University of Electronic Science and Technology of China (UESTC), Chengdu, China. In 2008, he joined the School of Electronic Engineering, UESTC. He is currently a Professor with the School of Information and Communication Engineering, UESTC. He has authored or coauthored more than 50 articles. His research interests include radar signal processing, synthetic aperture radar imaging, and target recognition.

Dr. Cao is a reviewer for several international journals and conferences, such as the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, the *IET Radar, Sonar and Navigation*, and *Remote Sensing*.

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